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Explicit-Implicit-Null Time Stepping: from Stiff, Nonlinear Anisotropic Diffusion to Linear Solves

MFEM Community Workshop, Georgia Tech, Atlanta, GA

September 25th, 2026



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Strategic Deterrence, Design Physics

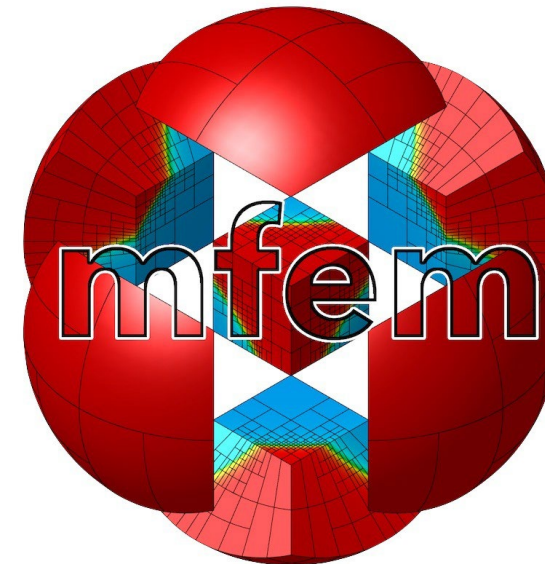


General Nonlinear Spatial Derivatives

Introduction of EIN

MOTIVATION: Temporal Bottlenecks

- Physics solvers like the **Modular Finite Element Methods (MFEM)** codebase typically combine spatial discretization with a time integrator.
- Temporal challenges of nonlinear spatial derivatives (Chacón et al., 2025):
 - Explicit methods: severely restricted by parabolic or worse CFL time-step limits
 - Implicit methods: must solve large, nonlinear, and expensive systems when physics are self-dependent.
- In systems, strong magnetic fields induce directional behavior (e.g., faster diffusion along field lines). We focus on **nonlinear anisotropic diffusion** as a model problem of temporal bottlenecks.



Stiffness: when the problem exhibits different time scales, requiring tiny time-steps to resolve the *fastest* time scale without instabilities

EIN and the IMEX Family

This raises the question: *can explicit and implicit methods be combined to improve performance?* e.g. compressible Navier-Stokes (Tan et al., 2025)

- Explicit–Implicit (IMEX) methods
 - Treat stiff (e.g., viscous) terms implicitly vs. non-stiff (e.g., inviscid) terms explicitly
 - First-order IMEX scheme: (Ascher et al., 1997)

$$u^{n+1} = u^n + \Delta t F(u^n) + \Delta t G(u^{n+1})$$

- **Explicit–Implicit–Null (EIN)** method
 - A special IMEX method for problems with stiff nonlinear spatial derivatives
 - Enables linear solves instead of nonlinear systems

EIN for Convection-Diffusion

- Using the method of lines, write the semi-discrete system as

$$u_t = F(t, u),$$

where $F(t, u)$ collects all spatial terms, including **stiff nonlinear derivatives**.

- To isolate stiffness, add and subtract a suitable “null” approximation G :

$$u_t = \underbrace{F(t, u) - G(t, u)}_{\mathcal{T}_1} + \underbrace{G(t, u)}_{\mathcal{T}_2}$$

Explicit (Non-Stiff) Implicit (Stiff, Linear)

- Goal:** Choose $G(t, u)$ to capture the stiff contributions in $F(t, u)$. Then, $\mathcal{T}_1$ is non-stiff (or at least less stiff), whereas $\mathcal{T}_2$ may be stiff and preferably linear. Apply standard IMEX: explicit on $\mathcal{T}_1$, implicit on $\mathcal{T}_2$ (Tan et al., 2025)

Choosing $G(t, u)$ in EIN

- The choice of $G(t, u)$ is flexible, as long as it captures the wave-number scaling of the stiff terms.

- Example from Wang et al. (2020): **nonlinear (isotropic) diffusion**

$$u_t = f(u)_x + (a(u)u_x)_x, \quad a(u) > 0$$

- Choose $G(t, u)$ as a linear diffusion approximation, leading to

$$u_t = \underbrace{f(u)_x + (a(u)u_x)_x}_{\mathcal{T}_1} - \underbrace{a_1 u_{xx}}_{\mathcal{T}_2} + \underbrace{a_1 u_{xx}}_{\mathcal{T}_2},$$

where a_1 is a constant diffusion coefficient chosen for stability.

- Apply standard IMEX: explicit on $\mathcal{T}_1$, implicit on $\mathcal{T}_2$.

- IMEX–EIN schemes of first, second, and third order have been developed and paired with various spatial discretizations (Wang et al., 2020; Tan et al., 2025).

- Stability requirements show stability constant is of the form

$$a_1 = a_0 \max(a(u)).$$

- First- and second-order IMEX: $a_0 \geq 0.5$
 - Third-order IMEX: $a_0 \geq 0.54$
- These conditions are proven via one-dimensional linear stability analysis. Numerical results also indicate stability and optimal accuracy persist in multi-dimensional, nonlinear settings.
- *Observation:* a_0 is optimal—smaller values fail under mesh refinement, while larger values remain stable but increase error.



Nonlinear Anisotropic Diffusion

Methodology

Anisotropic Diffusion

- Anisotropic thermal diffusion:

$$T_t = \nabla \cdot (\mathbf{D}(T) \nabla T),$$

where T is temperature, and $\mathbf{D}(T)$ is diffusion tensor.

- In 2D, the magnetic field direction is

$$\mathbf{b} = (\cos \alpha, \sin \alpha)^\top,$$

allowing for arbitrary alignment with the grid.

- Then,

$$\mathbf{D} = (D_{\parallel} - D_{\perp}) \mathbf{b} \mathbf{b}^\top + D_{\perp} \mathbf{I}$$

- Parallel diffusion coefficient $D_{\parallel}(T)$
- Perpendicular diffusion coefficient $D_{\perp}(T)$

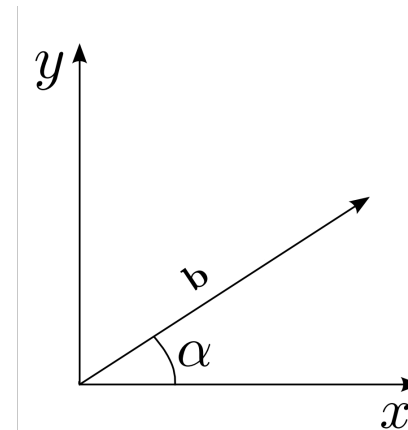


Figure from (van Es et al., 2014)

NOTE: coefficients are often temperature-dependent, introducing nonlinearity.

EIN for Anisotropic Diffusion

- Applying EIN,

$$T_t = \underbrace{\nabla \cdot (\mathbf{D}(T)\nabla T)}_{\text{Explicit } \mathcal{T}_1} - \underbrace{G(t, T)}_{\text{Implicit } \mathcal{T}_2}.$$

- The stiffness arises from the anisotropic diffusion operator. Choose $G(t, T)$ to encode anisotropy while remaining linear —a **weighted Laplacian**.
- In 2D, use appropriately large linear anisotropic diffusion

$$G(t, T) = a_1 \nabla \cdot (\mathbf{b}\mathbf{b}^T \nabla T) + a_2 \nabla \cdot (\mathbf{b}_\perp \mathbf{b}_\perp^T \nabla T),$$

where constants a_1 and a_2 represent stiff diffusion behavior.

Again, $a_1 = a_0 \max(D_\parallel(T))$ and $a_2 = a_0 \max(D_\perp(T))$ for stability factor a_0 .



MFEM Implementation

Numerical Results

Pure Isotropic & Anisotropic Diffusion

$$T_t = \nabla \cdot (a(T) \nabla T)$$

$$G(t, T) = a_0 \max(a(T)) \Delta T$$

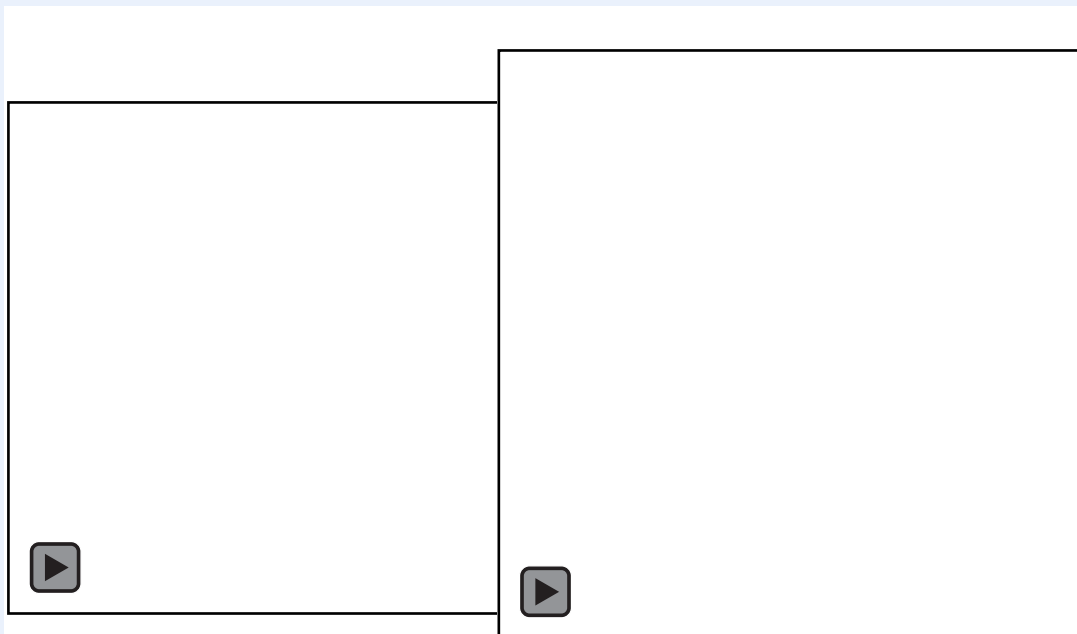


Figure 1a: Nonlinear isotropic diffusion of a symmetric Gaussian, nonlinear form of $a(T) = \kappa + \eta T$. (Left: surface, Right: contour) **Concentric symmetry remains.**

$$T_t = \nabla \cdot (\mathbf{D}(T) \nabla T)$$

$$G(t, T) = a_0 (\max(D_{\parallel}) \nabla \cdot (\mathbf{b}\mathbf{b}^T \nabla T) + \max(D_{\perp}) \nabla \cdot (\mathbf{b}_{\perp} \mathbf{b}_{\perp}^T \nabla T))$$

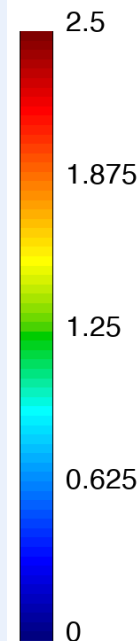
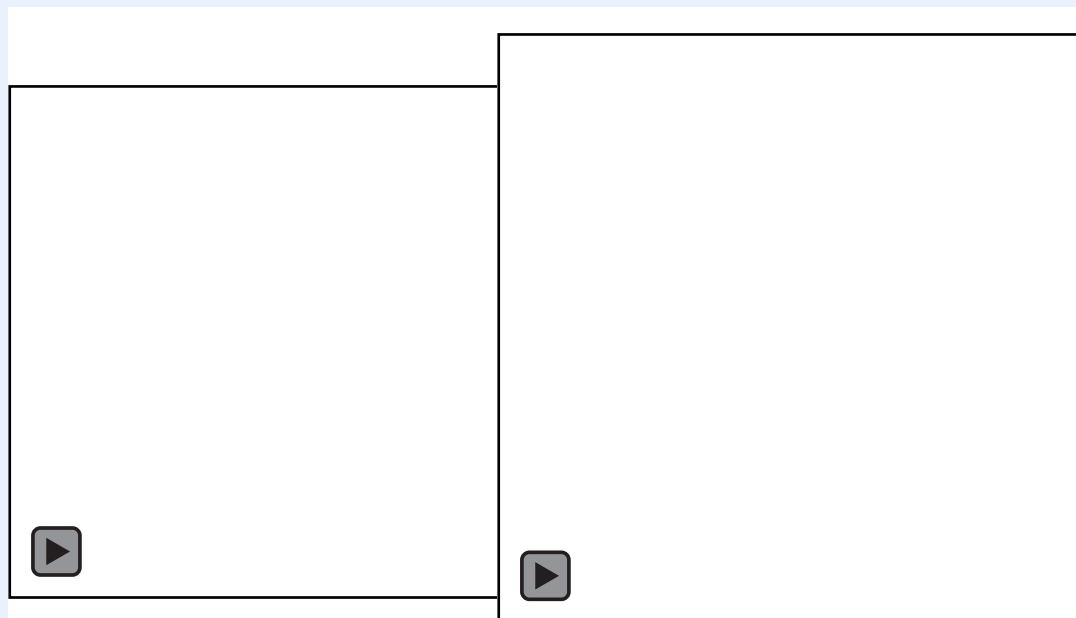


Figure 1b: Nonlinear anisotropic diffusion of a symmetric Gaussian, nonlinear form of $D_{\parallel}(T) = T^{2.5}$ with varying α . (Left: surface, Right: contour) **Strong anisotropy develops a ridge in parallel direction, rotating as α changes from 0° to 90° .**

High-Order Error Convergence via MMS

- Implemented isotropic and anisotropic diffusion, for various nonlinearities.
- Optimal orders are achieved for IMEX-EIN timesteppers of orders 1, 2, and 3.
- Two third-order schemes:
 - IMEX-DIRK3
 - IMEX-RK3

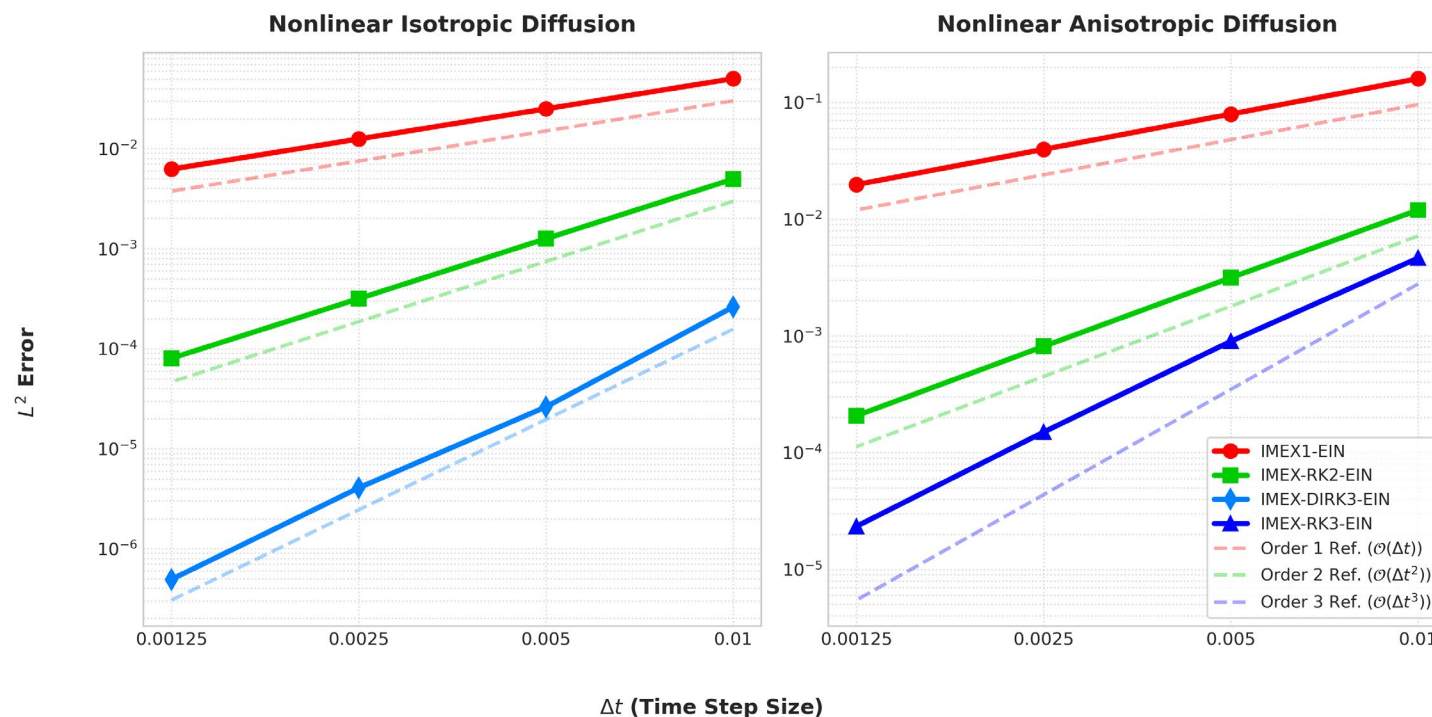


Figure 2. Error convergence for 2D problems. Left: nonlinear form of $a(T) = \kappa + \eta T$, Right: nonlinear form of $D_{||}(T) = T^{2.5}$ with time-varying α .

High-Order Performance Validation

- High-order schemes better condition systems and decrease error magnitude.
 - Even for cheap (and highly parallelizable) preconditioner `mfem::DSmooother`
- Reduction in iterations is a mathematical feature of the IMEX scheme's conditioning

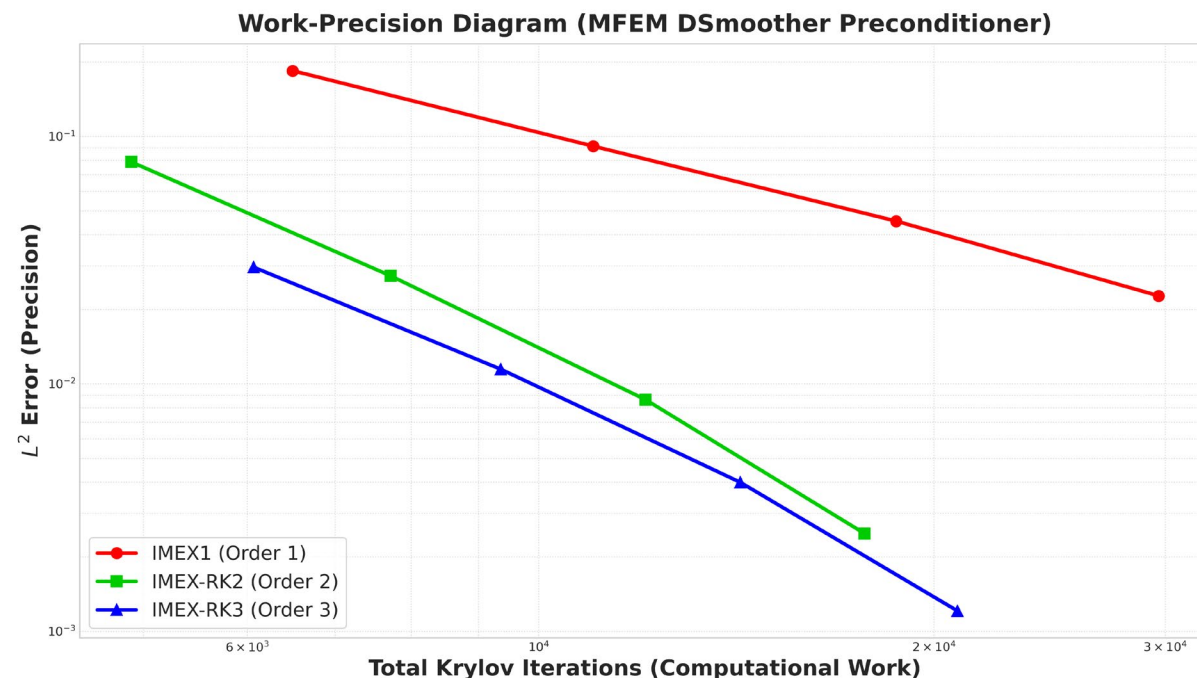


Figure 3. Work-precision trade-off among IMEX-EIN schemes for 2D anisotropic diffusion with nonlinear form of $D_{\parallel}(T) = T^{2.5}$ and time-varying α .

High-order methods are vital to performance scaling in scientific computing.

Coupling to Anisotropic Thermal Diffusion

- Electromagnetic Diffusion system, coupled via joule heating:

$$\begin{aligned}\nabla \times \nabla \times \mathbf{E} + \sigma(T)\mathbf{E} &= f \\ T_t - \nabla \cdot (\mathbf{D}(T)\nabla T) &= \mathbf{E} \cdot \sigma(T)\mathbf{E},\end{aligned}$$

where

- $\mathbf{E}$ is the electric field,
- σ is the electrical conductivity,
- f is a source term,
- T is temperature, and
- $\mathbf{D}$ is the thermal diffusion tensor, with diffusion coefficients $D_{\parallel}(T)$ and $D_{\perp}(T)$.

- At timestep t^n ,
 - Given T^n , solve for $\mathbf{E}^n$
 - Update joule heating $Q^n = \mathbf{E}^n \cdot \sigma(T^n)\mathbf{E}^n$
 - Solve for T^{n+1} , via EIM:

$$T_t^{n+1} = \underbrace{\nabla \cdot (\mathbf{D}(T^n)\nabla T^n) + Q^n - G(T^n, t)}_{\text{Explicit } \mathcal{T}_1} + \underbrace{G(T^{n+1}, t)}_{\text{Implicit } \mathcal{T}_2}$$

High-Order Error Convergence via MMS

- Preliminary work for linear and nonlinear thermal diffusion, with constant angle of misalignment.
- MFEM capabilities:
 - $\mathbf{E} \in H(\text{curl})$ and $T \in H^1$
- Optimal orders are achieved for $\mathbf{E}$ and T using IMEX-EIN timesteppers of orders 1, 2, and 3.
 - $\mathbf{E}$ is steady-state but coupled to transient T

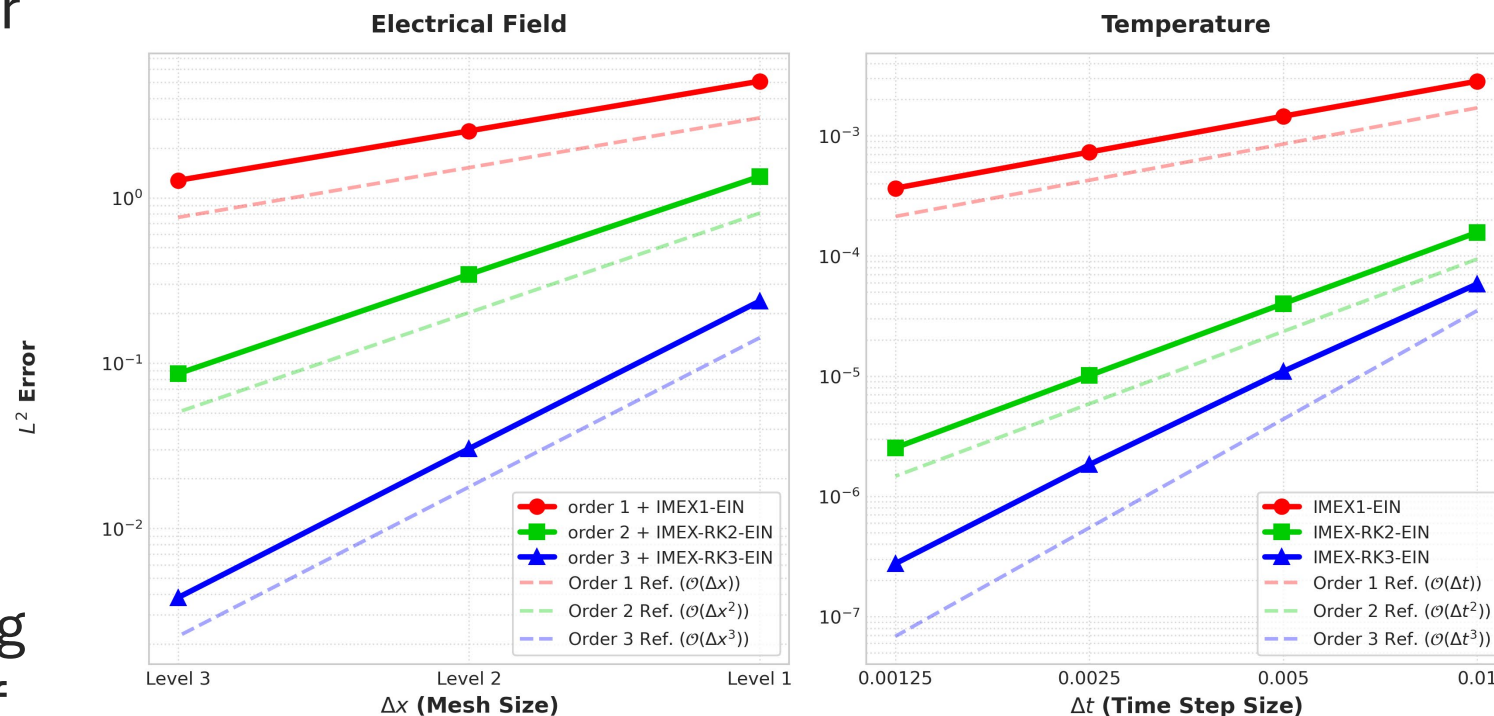


Figure 4. EM-thermal diffusion via joule heating coupling, with nonlinear anisotropy $D_{\parallel}(T) = T^{2.5}$ and constant α .

High-Order Performance Validation

- Higher-order IMEX methods achieve much smaller error magnitude.
- High accuracy comes at increased total solver cost because multistage methods require more EM and thermal solves.

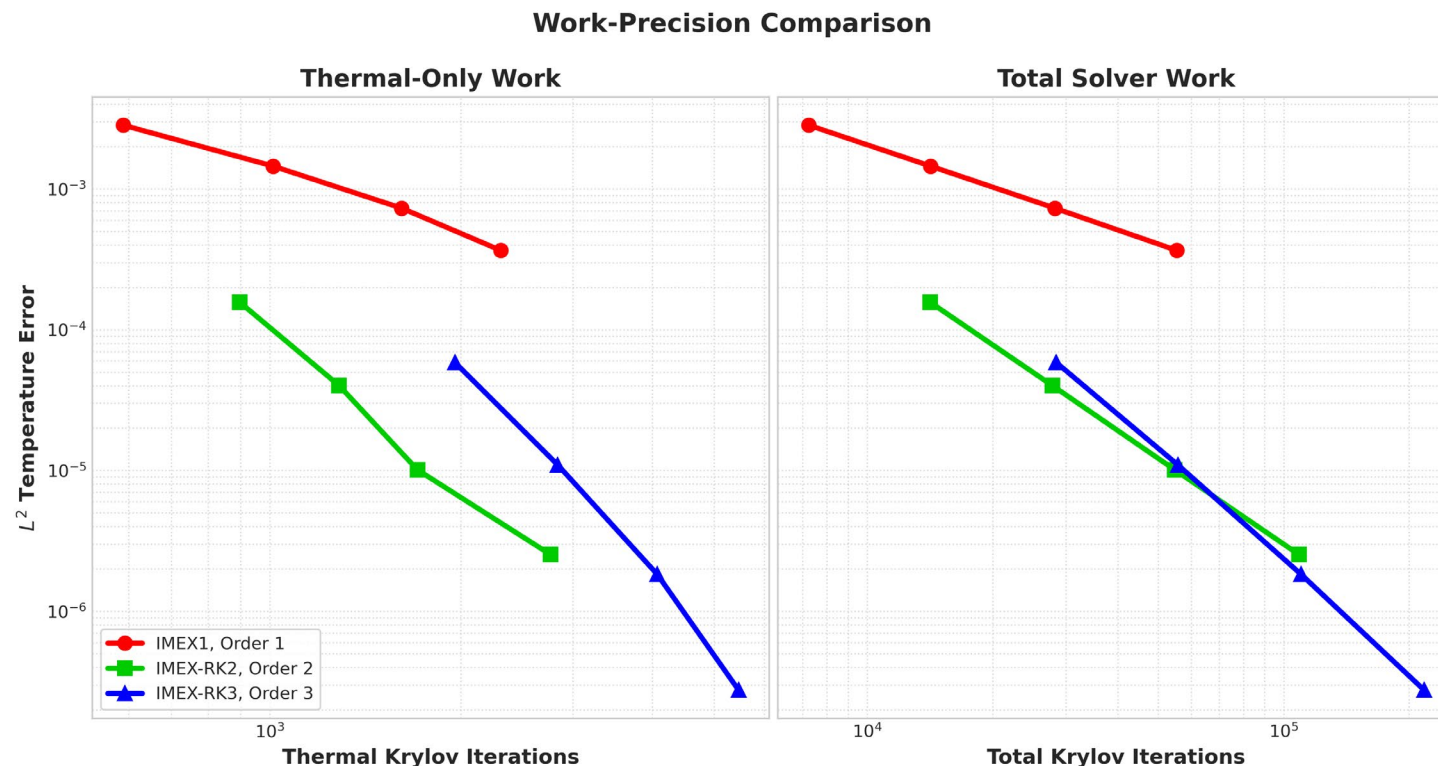


Figure 5. Work-precision trade-off among IMEX-EIN schemes. Left: only anisotropic thermal diffusion, Right: full EM-thermal coupling (both EM + thermal iterations).

High-order methods are vital to performance scaling in scientific computing.



Conclusions

Conclusions

Explicit–Implicit–Null (EIN) framework extends naturally to nonlinear anisotropic diffusion in physical systems.

- **Novel:** transforms stiff, nonlinear problems into stable, linear ones.
- **Robustness:** handles misalignment and various nonlinearities.
- **High-Order Value:** demonstrates how high-order methods are vital to computational speedups in scientific computing.

Future Work: coupling anisotropic thermal diffusion within larger physical systems in MFEM; performance analysis under strong anisotropy.

EIN enables stable and efficient time integration — a promising building block for complicated physics.



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